Competitive Exam Practice 3: System Calls

GATE (Computer Science) Pattern

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October 17, 2026

Concepts That Click

Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type. *All questions in this set are original, written in the GATE pattern; none is claimed to be a past-year question.*

Q1 *[Original, GATE-pattern]* **[MCQ, 1 mark]** Which of the following statements about the relationship between a library function and a system call is **TRUE**?

(A) Every library function that performs I/O executes exactly one system call

(B) A library function may complete entirely in user mode and issue no system call at all

(C) A system call executes in user mode but is allowed to read kernel data

(D) `printf` is a system call because it eventually produces output

Q2 *[Original, GATE-pattern]* **[MCQ, 1 mark]** After the trap instruction transfers control to the kernel, the system-call number supplied by the library stub is used to:

(A) search the table by comparing the name of the requested service

(B) index the system-call table, whose entry is a pointer to the handler

(C) place the address of the handler in the calling process's page table

(D) select the privilege level at which the handler will execute

Q3 *[Original, GATE-pattern]* **[MCQ, 1 mark]** A kernel cannot renumber its existing system calls without breaking already-compiled programs. The most direct reason is that:

(A) the numbers are stored in the CPU's control registers at boot time

(B) the numbers are part of the ABI and are embedded in compiled binaries

(C) the trap instruction can carry only an 8-bit number

(D) renumbering would change the calling convention used for arguments

Q4 *[Original, GATE-pattern]* **[MCQ, 2 marks]** A system-call ABI reserves six registers for arguments and uses a separate register for the system-call number. A new service needs nine scalar arguments. Which mechanism is most appropriate, and why?

(A) Registers: the nine arguments can be packed into the six reserved registers

(B) A block or table in memory: the nine arguments are stored contiguously and the block's address is passed in one register

(C) The stack: it is always faster than a memory block because the kernel never copies anything

(D) None: a service with more arguments than argument registers cannot be implemented on this ABI

Q5 *[Original, GATE-pattern]* **[MCQ, 2 marks]** A user process passes to `read` a pointer whose value lies in the kernel's address range. What does a correctly designed kernel do?

(A) Dereference it, because the kernel is permitted to access all physical memory

(B) Validate the range against the process's address space and fail the call with `EFAULT`

(C) Kill the process immediately with `SIGSEGV`, without returning to user mode

(D) Copy the kernel data into the user buffer, since `read` does not modify the source

Q6 *[Original, GATE-pattern]* **[NAT, 2 marks]** A system-call ABI reserves six registers for integer arguments and places the system-call number in a separate register. A service takes eleven integer arguments, each 8 bytes. The first six travel in registers; the remaining five are stored contiguously in a memory block whose address is passed in one register. The number of bytes of argument data that the kernel must copy from the user's block is _____.

Q7 *[Original, GATE-pattern]* **[NAT, 2 marks]** A shell runs `cp` on a file of 96 KiB; `cp` uses an 8 KiB buffer and the file size is an exact multiple of the buffer size. The shell issues one `fork`, the child issues one `exec`, and the parent issues one `wait`. After `exec`, `cp` issues two `open` calls, one `read` and one `write` for every full buffer, two `close` calls, and one `exit`. Counting only the calls named here, the total number of system calls issued by the shell and its child for this command is _____.

Q8 *[Original, GATE-pattern]* **[NAT, 2 marks]** A program copies a file of 1 MiB. Version A uses a 4 KiB buffer and issues one `read` and one `write` per buffer; Version B issues a single `read` and a single `write` for the entire file. Both versions move the same number of bytes. The factor by which Version A increases the number of user-to-kernel transitions relative to Version B is _____.

Q9 *[Original, GATE-pattern]* **[NAT, 2 marks]** Consider the following system calls:

`fork`, `open`, `ioctl`, `getpid`, `socket`, `chmod`, `lseek`, `wait`, `close`, `send`.

The number of these calls that belong to the *file-management* category is _____.

Q10 *[Original, GATE-pattern]* **[NAT, 2 marks]** A kernel's system-call table has 340 entries, numbered 0 through 339. Twelve new calls are appended to the table. Five obsolete calls are withdrawn from service, but their table entries and numbers are retained for compatibility. The largest valid system-call number after these changes is _____.

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